
NETWORK MODELLING TO ESTIMATE REPRODUCTION NUMBERS

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ABSTRACT

Human movement can initiate, sustain, and prolong outbreaks of both directly transmitted and vector-borne diseases, yet our real-time operational tools for estimating the instantaneous reproduction number $R(t)$ often assume static, homogeneously mixing populations and do not capture when and where individuals transmit according to different conditions based on their movement. Renewal equations underpin many such popular estimators of $R(t)$, and here, we develop a network-based model from which we derive new mechanism-led renewal equations for both directly transmitted and vector-borne diseases. These equations directly integrate within-day human movement to rigorously define inward and outward $R(t)$ for each individual locations, $R(t)$ between locations, $R(t)$ at meeting locations, and $R(t)$ for the entire mobility network of locations. These quantities provide a required new meaning for estimating reproduction numbers of and controlling growth of real-world outbreaks that are shaped by humans moving across connected locations. We demonstrate how our new framework's outputs can be used alongside mobile phone data to design targeted policy interventions, including the strength, type, and length of intervention required across space and time. We further show how to explore short-term perturbations and transience in the epidemic and how standard estimators bias our estimates when within-day human movement is ignored. We illustrate our framework with applications to epidemics on different types of city networks.

1 Introduction

For diseases that affect humans, human movement plays a crucial role in pathogen spread, irrespective of the route of transmission (direct, sexual, or vector-borne). Human movement enables introductions, continued establishment, persistence, spatial spread, and reintroductions^[1,2,3,4].

In recent years, there has been an increasing volume of human movement data being available to researchers^[5], and with this, an increasing focus of the role of human movement on epidemic dynamics^[4,6,7,8,9]. Indeed, there is an increasingly wider variety of the types of human mobility data that are commonly used by researchers, each of which have different strengths and limitations based on different research/policy questions and the local context^[7,10,11].

Despite the essential role of humans and the growing availability of data, the relative roles of human movement are rarely integrated in the most common methods for estimating real-time transmissibility (reproduction numbers $R(t)$) or epidemic speed (growth rates $r(t)$). Instead, multi-patch models (e.g. metapopulation or network models) are often used to understand how human movement impacts outbreaks across a mobility network^[12], while simpler single-patch models (e.g. renewal-equation-based models) are the basis for estimating $R(t)$ and $r(t)$ for individual locations.

While multi-patch models can offer greater expressiveness, this can come at the expense of increased complexity and hinder real-time inference. This differs to traditional renewal equations for directly transmitted diseases and analogous equations for vector-borne disease, where a flexible yet very general model allows for fast estimating of $R(t)$ with few assumptions. However, key limitations of the renewal equation include homogeneous mixing of individuals and the estimation of quantities such as $R(t)$ for each location independently. Hierarchical models using renewal equations could borrow information/statistical strength across locations, yet these fail to capture the true mechanistic meaning of $R(t)$. This is not just a mathematically limiting factor but also potentially misleading as the local transmissibility can be over/under-estimated, human movement effects ignored, and public health policy misinformed, or at least not fully informed^[13].

Recent approaches^[14,15] adapt the renewal equation or similar frameworks to incorporate human movement into estimates of $R(t)$. Other common statistical approaches allow for a background/external term, say $m(t)$, which creates two competing processes; one that accounts for infection importations and another that accounts for infections generated locally/internally^[16,17,18,19]. Each of these approaches are causally insufficient for describing how infected individuals generate new infections at differing rates across space and time based on the movement patterns and contact rates of susceptible and infected individuals, the depletion of susceptible populations, and the biological infectiousness of infected individuals.

Here, we have similar objectives, yet take a slightly different approach, as we return to first principles and develop a spatially explicit and mobility-informed mechanistic model that is described by a set of PDEs, from which we derive mechanism-led renewal equations. Our resulting renewal equations differ to existing approaches^[14,15] in four ways; i) we derive reproduction numbers across the network using the required mechanistic formula (that integrates the instantaneous kernel), ii) we derive a wide taxonomy of rigorously defined reproduction numbers, kernels, and generation time distributions for all pairwise combinations of locations, for inward and outward transmission, for meeting locations, and for the epidemic network as a whole, iii) we integrate within-day human movement dynamics to precisely measure all infection opportunities for an individual who may spend fractions of their day in one, two, or more locations, and iv) we extend our new approach to mobility-informed renewal equations for vector-borne diseases and diseases with time-varying generation distributions. Our mechanistic approach provides a wide array of new, policy-relevant quantities across space and time that have been unachievable from existing $R(t)$ estimation methods, which can be used to determine the strength, type, and duration of interventions required to control an outbreak.

2 Materials and methods

2.1 Reproduction numbers, kernels, and generation times

During an outbreak, to respond and plan appropriately, decision-makers need timely and reliable information about how fast a pathogen is spreading at any time t . Reproduction numbers are unitless quantities that are used to measure epidemic strength, which informally are how many infections are generated by an “average individual”. These mathematical, counterfactual indicators have become popular and useful ways to study and quantify pathogen invasion, epidemic growth, population immunity^[20,21,22,23]. In particular, the threshold property of values greater/less than one is used to indicate whether long-term epidemic growth is possible.

At the start of an outbreak, the basic reproduction number, R_0 , for a specific pathogen and population, defines the average number of individuals infected by an average primary individual in a fully susceptible population over their infected lifetime^[20,21,24,25]. As an outbreak unfolds, the susceptible population often depletes due to naturally acquired

immunity, vaccination, or deaths, and the instantaneous reproduction number $R(t)$ can be used to quantify how transmissibility changes over time^[19,26,27]. For a single location, $R(t)$ is defined for each time t as the expected number of new human infections generated by an infected individual throughout the course of their infection, should the conditions affecting human transmission at time t remain the same. This human-to-human reproduction number can be defined for directly transmitted and vector-borne diseases alike, and mathematically is derived by integrating over the instantaneous kernel $K(t, \tau)$:

$$R(t) = \int_0^t K(t, \tau) d\tau,$$

where $K(t, \tau)$ represents, at time t , the contribution to new human infections by an individual of infection age τ . Then, the probability density for the generation time distribution (which can vary over calendar time) is defined as:

$$g(t, \tau) = \frac{K(t, \tau)}{R(t)},$$

thereby representing the relative contribution to new human infections at time t from individuals of infection age τ . Alternatively, this can be thought of as the instantaneous distribution of times between primary and secondary human infections.

R_0 and $R(t)$ are also often derived by using a structured population model and computing the spectral radius $\rho(\mathbf{K})$ of the next-generation operator (NGO) $\mathbf{K}$ ^[24,28,29,30,31,32]:

$$R(t) := \rho(\mathbf{K}), \quad (1)$$

where the NGO $\mathbf{K}$ is a positive operator that represents the number of new infections in the next generation of transmission (the wave of secondary infection). If there is a finite number of states, the NGO $\mathbf{K}$ becomes the next-generation matrix (NGM) and $R(t)$ is simply the largest real absolute eigenvalue of $\mathbf{K}$.

Again, similar results hold for directly transmitted and vector-borne diseases^[33,34], and for heterogeneous populations^[35]. However, all of the above implicitly assumes a single-patch model which defines how past human infections generate now human infections in that location.

2.2 Renewal equations

The renewal equation is a natural way to capture how past human infections generate now human infections with delay, and works by combining $R(t)$ and $g(t, \tau)$ with past human infections. It was first introduced by... History: Euler, Lotka, McKendrick More recently,^[36] Extended for vector-borne^[34], heterogeneous populations^[35,37].

While there are many ways to estimate $R(t)$ from data in practice^[32,38,39], we will focus on developing the necessary mechanistic machinery for defining $R(t)$ in connected populations.

Practically, there are many ways to estimate $R(t)$ from data, yet the renewal equation is the basis for many popular software tools that estimate $R(t)$ and additional epidemiological quantities, including EpiEstim^[26], EpiNow2^[40],^[26,40,41]. It is most often used in discrete time to align with data.

2.3 Deriving renewal equations from age-structured models

In^[34], Here, we provide a similar derivation for directly transmitted, from which a renewal equation can be derived.

2.4 Human mobility data

To embed human movement in our model, we require a human mobility data source that can account for within-day movements between locations. A natural source is mobile phone data. Similar to^{[?]1}

- Typical: Origin-destination matrix
- Lim of flow data; doesn't account for proportion of day spent in infection-opportunity-generating settings -> over-estimate infection opportunity/risk throughout day
- What we want is something that measures exposure risk....
- Two example ways; mobile phone ping data and co-location data, mobile phone ping data measures fraction of day spent in individual locations, co-location data can processed to obtained similar estimates.
- Alternatives: Show how bad gravity + radiation model approximations are

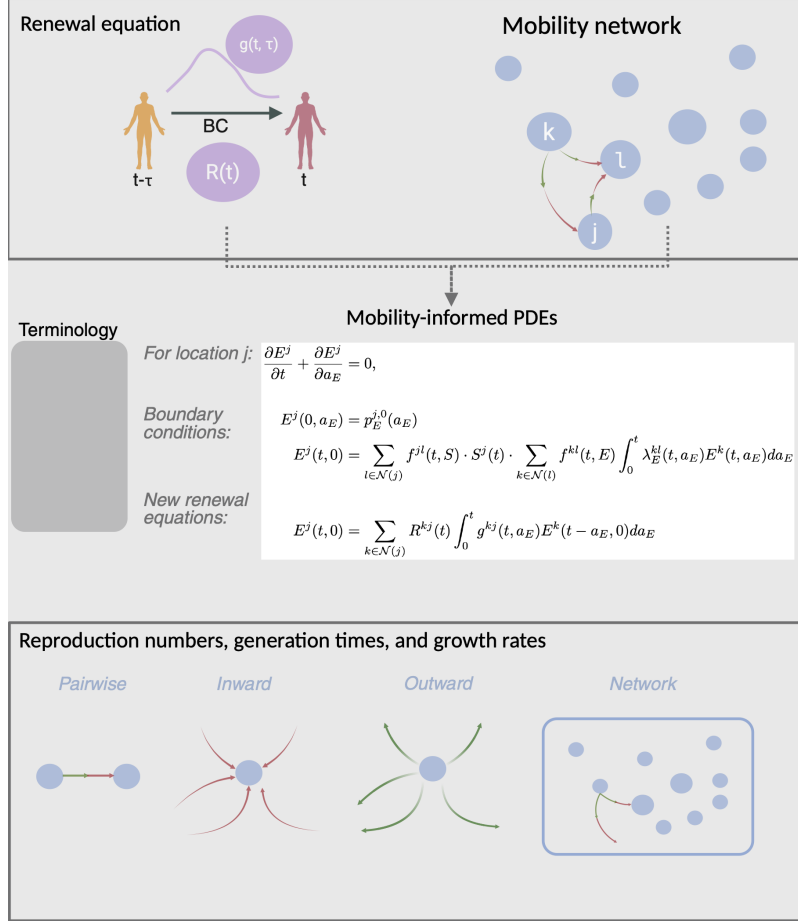


Figure 1: **Schematic of key concepts:** To include, clearer schematic, GT (what various types mean), existing approaches?

2.5 Model implementation and analysis

- Define numerical simulation routine (discretisation) in Python.
- For directly transmitted, assume SARS-CoV-2-like epidemiological parameters
- Independent $R(t)$ estimation:
- Sensitivity and elasticity: System-level $\mathcal{R}(t)$ to individual elements $R^{jk}(t)$ (computed from matrix perturbation theory), to mobility changes (fraction of time, number of trips), and to parameters

3 Results

We now extend the renewal equation to encompass a human mobility network that allows for movement of and susceptible infected individuals, thereby allowing for all within-day infection opportunities in an epidemic on a network.

3.1 Mobility-informed PDEs, renewal equations and reproduction numbers for directly transmitted diseases

3.1.1 A new network modelling framework

Suppose that we have a mobility network that contains $\mathcal{L}$ locations. For a directly transmitted pathogen such as SARS-CoV-2, we can define a mobility-informed system of age-structured, conservation PDEs as:

$$\frac{\partial E^j}{\partial t} + \frac{\partial E^j}{\partial a_E} = 0, \quad (2)$$

where $E^j(t, a_E)$ denotes the density of individual humans resident in location j at time t who were exposed to a pathogen a_E time units ago and became immediately infectious. a_E therefore represents an individual's infection age, not biological age, or equivalently their time since state entry.

We close the system with boundary conditions, thereby introducing mobility effects via:

$$E^j(0, a_E) = p_E^{j,0}(a_E) \quad (3)$$

$$E^j(t, 0) = \sum_{l \in \mathcal{N}(j)} \underbrace{f^{jl}(t, S) \cdot S^j(t)}_{(a)} \cdot \sum_{k \in \mathcal{N}(l)} \underbrace{f^{kl}(t, E) \int_0^t \lambda_E^{kl}(t, a_E) E^k(t, a_E) da_E}_{(b)} \quad (4)$$

where $p_E^{j,0}(a_E)$ denotes the initial density of infected humans, $f^{jk}(t, S)$ and $f^{jk}(t, E)$ denote the probabilities respectively that a susceptible and exposed individual from location j is in location k at time t with $\sum_{k \in \mathcal{N}(j)} f^{jk}(t, S) = 1$ and $\sum_{k \in \mathcal{N}(j)} f^{jk}(t, E) = 1$ for all locations j , and $\lambda_E^{kl}(t, a_E)$ is the transmission rate, inclusive of recovery probability, for an infected individual of infection age a_E resident in location k but (potentially) transmitting in meeting location l . (a) accounts for the active population of susceptible individuals resident in location j that are present in location l . (b) accounts for infected individuals that are residents in location k transmitting to individuals present in location l . In words, the boundary condition allows for susceptible residents of location j to be infected in any location l by infected individuals resident in any location k .

$f^{jl}(t)$ is most naturally processed from the fraction of time (e.g. daily) that an individual resident in location j spends in location l . Alternatively, we could write the model using the probability that an individual from location j meets an individual from location k , yet this would not account for within-day human mobility as accurately. $f^{jl}(t, S)$ and $f^{jl}(t, E)$ could be different for susceptible versus infected individuals and for different times (both within days and over several days). $f^{jl}(t, E)$ could also be a function of infection age if we suspect that disease symptoms or test diagnosis affect human movement, and/or a demographic such as biological age or occupation. $f^{jl}(t, S)$ could be a function of epidemic stage, concurrent public health policy (e.g. mobility reductions), or public awareness (e.g. during information campaigns).

$\lambda_E^{kl}(t, a_E)$ could be different for pairwise combinations of locations and possibly a function of the activity (e.g. numbers/frequency of within-location trips) experienced by a resident of location k in location l . We could have $\lambda_E^{kk}(t, a_E) > \lambda_E^{kl}(t, a_E)$ for $k \neq l$. This could reflect transmission being more likely within the home locations due to the nature of contacts experienced there (e.g. household versus workplace). Further realism could be incorporated by not just conditioning on whether the infector is in their home residence, but also whether the putative infectee (i.e. the susceptible individual) is in their home residence. This can remove the assumption of homogeneous mixing within each meeting location such that we have $\lambda_E^{kk}(t, a_E | S^k) \neq \lambda_E^{kl}(t, a_E | S^l)$ for $l \neq k$.

What would $\lambda_E^{kl}(t, a_E)$ look like for SARS-CoV-2 and other directly transmitted infectious diseases? The transmission rate $\lambda_E^{kl}(t, a_E) = \frac{1}{N_{\text{eff}}^l(t)} \times \kappa^{kl}(t) \times p^k(a_E)$ is the (potentially) time-varying contact rate times the probability that the contact results in an infection, scaled by the total population $N_{\text{eff}}^l(t) = \sum_{q \in \mathcal{N}(l)} f^{ql}(t) N^q(t)$ in location l at time t (for frequency-dependent transmission). $\kappa^{kl}(t)$ could be a function of the number of (within-location) flows taken by the resident of location k in location l , while $p^k(a_E)$ could be based on infectiousness profiles and potentially vary based on demographics of residents of location k .

3.1.2 Deriving renewal equations and reproduction numbers

Using the method of characteristics and assuming longer-term dynamics (i.e. $t > a_E$), the boundary condition can be written as a renewal equation that describes how past human infections generate new human infections:

$$E^j(t, 0) = \sum_{l \in \mathcal{N}(j)} f^{jl}(t, S) \cdot S^j(t) \cdot \sum_{k \in \mathcal{N}(l)} f^{kl}(t, E) \int_0^t \lambda_E^{kl}(t, a_E) E^k(t - a_E, 0) da_E. \quad (5)$$

As we have finite sums and the integrand is non-negative, we can apply Tonelli's Theorem to rearrange the sums to isolate the infector's resident location k , we have:

$$E^j(t, 0) = \sum_{k \in \mathcal{N}(l)} \int_0^t \left[\sum_{l \in \mathcal{N}(k, j)} f^{jl}(t, S) \cdot S^j(t) \cdot f^{kl}(t, E) \lambda_E^{kl}(t, a_E) \right] E^k(t - a_E, 0) da_E. \quad (6)$$

where we use shorthand $\mathcal{N}(k, j) := \mathcal{N}(k) \cup \mathcal{N}(j)$.

In matrix notation, we can write this as:

$$\mathbf{E}(t, 0) = \int_0^t \mathcal{K}(t, a) \mathbf{E}(t - a_E, 0) da_E, \quad (7)$$

or

$$(\mathbf{E}(t, 0))_j = \sum_{k \in \mathcal{N}(l)} \int_0^t \mathcal{K}_{jk}(t, a_E) E^k(t - a_E, 0) da_E, \quad (8)$$

where $\mathcal{K}_{jk}(t, a_E) := \mathbb{K}^{kj}(t, a_E)$, i.e. the instantaneous kernel at each time t that shapes how infected residents of location k of different infection ages a_E generate new infections in residents of location j (which can occur in any meeting location):

$$\mathbb{K}^{kj}(t, a_E) := \sum_{l \in \mathcal{N}(j, k)} f^{jl}(t, S) \cdot S^j(t) \cdot f^{kl}(t, E) \lambda_E^{kl}(t, a_E), \quad (9)$$

$$\mathbb{K}_{out}^k(t, a_E) := \sum_{j \in \mathcal{N}(k)} \sum_{l \in \mathcal{N}(j, k)} f^{jl}(t, S) \cdot S^j(t) \cdot f^{kl}(t, E) \lambda_E^{kl}(t, a_E), \quad (10)$$

$$(11)$$

while $\mathbb{K}_{out}^k(t, a_E)$ defines the instantaneous rate at which infected residents of location k of infection age a_E generate new infections in residents of any location. These instantaneous kernels will depend on day-to-day (and diurnal if using sub-daily data) movement patterns as during-week (and during-day) infection dynamics may be more strongly affected by imported infections, while during-weekend (and during-night) dynamics may be more skewed towards within-location resident-to-resident transmission.

By integrating the instantaneous kernels over all infection ages and summing over locations, we recover different types of instantaneous reproduction numbers across space and time^[33,36]:

$$R^{kj}(t) := \int_0^t \mathbb{K}^{kj}(t, a_E) da_E \quad (12)$$

$$R_{out}^k(t) := \sum_{j \in \mathcal{N}(k)} R^{kj}(t) = \sum_{j \in \mathcal{N}(k)} \int_0^t \mathbb{K}^{kj}(t, a_E) da_E. \quad (13)$$

The *between-location/pairwise* $R^{kj}(t)$ is an instantaneous reproduction number as it is the expected number of new human infections generated by individuals resident in location k in individuals resident in j throughout the course of the infected individual's lifetime, if the conditions affecting transmission from residents of location k to residents of location j were to remain the same as at time t . $R_{out}^k(t)$ are the *outward* reproduction numbers defining the expected number of new human infections generated by an infected resident of location k in any location throughout their infected lifetime, should conditions affecting outward transmission from location k remain the same as at time t . $R^{kj}(t)$ can track the pairwise transmissibility patterns necessary for between- and within-neighbour transmission. $R_{out}^k(t)$ can quantify whether a given location is a net exporter of transmission (i.e., can infected residents sustain transmission in all locations) as it has the familiar threshold property of instantaneous, per-generation growth/decay at time t , but for the outbreak of secondary infections in any individual that results specifically from infected residents of location k .

As usual^[33], we can derive probability densities for the generation time distributions, at each calendar time, by dividing the instantaneous kernel by the corresponding reproduction number. Here, we have several different types of generation time distributions, each corresponding to different types of *generations* across space and time:

$$g^{kj}(t, a_E) = \frac{\mathbb{K}^{kj}(t, a_E)}{\int_0^t \mathbb{K}^{kj}(t, a_E) da_E} = \frac{\mathbb{K}^{kj}(t, a_E)}{R^{kj}(t)}, \quad (14)$$

$$g_{out}^k(t, a_E) = \frac{\sum_{j \in \mathcal{N}(k)} \mathbb{K}^{kj}(t, a_E)}{\sum_{j \in \mathcal{N}(k)} \int_0^t \mathbb{K}^{kj}(t, a_E) da_E} = \frac{\mathbb{K}_{out}^k(t, a_E)}{R_{out}^k(t)}, \quad (15)$$

where $g^{kj}(t, a_E)$ defines, instantaneously at time t , the relative contribution to the number of new human infections generated in residents of location j by humans resident in location k that were infected a_E time units ago. The shape of this distribution will be determined by the co-presence of the infected and susceptible residents of locations k and j respectively, their contact rates, and infectiousness profile. $g_{out}^k(t, a_E)$ defines, at time t , the relative contribution to the number of new human infections generated in residents of any location caused by humans resident in location k

that were infected a_E time units ago. The shape of this distribution will depend on the movement patterns, contact rates, and infectiousness profiles of the infected individuals in location k as they generate secondary infections in any location with time-varying totals of susceptible individuals. Therefore, the shape of these generation times will depend on biological infectiousness profiles, contact rates, susceptible and infected population counts, and importantly, the fraction of time spent by residents in different locations and whether this coincides with susceptible/infected individuals either staying in or moving to the meeting location. This is similar to the results of^[33], but with the addition of the spatial dimension to allow for interacting populations.

We can define our renewal equation per location in a more familiar way, capturing how previously infected individuals generate new infections in location j :

$$E^j(t, 0) = \sum_{k \in \mathcal{N}(j)} \int_0^t \mathbb{K}^{kj}(t, a_E) E^k(t - a_E, 0) da_E \quad (16)$$

$$= \sum_{k \in \mathcal{N}(j)} R^{kj}(t) \int_0^t g^{kj}(t, a_E) E^k(t - a_E, 0) da_E. \quad (17)$$

If we set $f^{jk}(t, S) = \delta_{jk}$ and $f^{jk}(t, E) = \delta_{jk}$ for all locations j and k , then we recover the familiar single-patch renewal equation for a closed population without importations:

$$E^j(t, 0) = \int_0^t \mathbb{K}^{jj}(t, a_E) E^j(t - a_E, 0) da_E \quad (18)$$

$$= R^{jj}(t) \int_0^t g^{jj}(t, a_E) E^j(t - a_E, 0) da_E. \quad (19)$$

Note that $R^{jj}(t)$ may differ from existing estimators of local $R(t)$ in closed populations, as $R^{jj}(t)$ accounts for outward movement of susceptible and infected residents from location j , as well as the inward movement of susceptible residents from other locations which may reduce transmissibility for frequency-dependent transmission.

3.1.3 Asymptotic and transient epidemic dynamics

We can assemble our between-location reproduction numbers into a valid, spatially structured NGM, using the convention of rows as infectees and columns as infectors to align with previous work^[13,28,30,31,32]:

$$\mathbf{R}(\mathbf{t}) = \begin{pmatrix} R^{11}(t) & R^{21}(t) & \dots & R^{\mathcal{L}1}(t) \\ R^{12}(t) & \dots & \dots & R^{\mathcal{L}2}(t) \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & R^{\mathcal{L}\mathcal{L}}(t) \end{pmatrix},$$

from which the system-/network-level reproduction number can be derived as the spectral radius of this reproduction/NGM^[13,28]:

$$\mathcal{R}(t) := \rho(\mathbf{R}(\mathbf{t})). \quad (20)$$

This instantaneous measure defines the asymptotic growth/decay of the linearised per-generation epidemic on the network. This is because we can apply the Perron-Frobenius Theorem to ensure a unique, dominant eigenvalue $\rho(\mathbf{R}(\mathbf{t}))$ that is simple, real and positive. We can apply the theorem as $\mathbf{R}(t)$ is a non-negative, irreducible matrix. These conditions are satisfied as $f^{jk}(t, S) \geq 0$ for all location pairs (j, k) , $S^j(t) \geq 0$ for all locations j , $f^{kl}(t, E) \geq 0$ for all location pairs (k, l) , and $\lambda_E^{kl}(t, a_E) \geq 0$ for all location pairs (k, l) .

Above, we align with previous work^[13,24,28] and define $\mathbf{R}(\mathbf{t})$ with rows corresponding to infectees and columns corresponding to infectors, but the spectral radius (and hence, $\mathcal{R}(t)$), will be the same if defined with rows corresponding to infectors and columns corresponding to infectees. To capture the heterogeneity in inward (e.g. high-burden locations) or outward (e.g. superspreading locations) transmission, we can compute the coefficient of variation (CV) of the row or column sums respectively of $\mathbf{R}(t)$.

Note that $\mathcal{R}(t)$ is *not* the expected/average number of secondary infections per infected human throughout their infected lifetime, should transmission conditions remain the same as at time t , as this will be quantified per location by $R_{out}^j(t)$ and depend on the residence location j of the infected human. Instead, applying $\mathbf{R}(t)$ to the incidence vector $\mathbf{E}(t, 0)$ repeatedly for many generations n , it will grow/decay as $(\mathcal{R}(t))^n$. $\mathcal{R}(t)$ therefore, unlike the pairwise $R_{kj}(t)$, provides a threshold property for the network, with values of $\mathcal{R}(t) > 1$ indicating asymptotic growth everywhere on the network and values of $\mathcal{R}(t) < 1$ indicating asymptotic decay everywhere on the network.

Indeed, the eigenvalue problem definition (Section A and^[13]) can provide further intuition as the right eigenvector $\mathbf{v}^*$ (from $\mathbf{R}\mathbf{v}^* = \mathcal{R}(t)\mathbf{v}^*$) defines the long-term, equilibrium spatial distribution of new infections where v_j^* is the fraction of infections that would occur in residents of location j .

$\mathcal{R}(t)$ only provides an instantaneous (counterfactual, asymptotic) measure of epidemic growth/decay, yet we can also explore transience of the epidemic. The subdominant eigenvalue $\lambda_2(t)$ governs the slowest-decaying transient mode, can be real or complex (if $\mathbf{R}(t)$ is asymmetric), and importantly, controls how quickly the epidemic converges to this asymptotic spatial distribution $\mathbf{v}^*$.

The mixing time ratio is defined as:

$$s(t) := \frac{|\lambda_2(t)|}{\mathcal{R}(t)} \quad (21)$$

measures instantaneously the mixing of the epidemic on the network, i.e. how many generations it takes for the spatial distribution of infections to stabilise. Values of $s(t)$ close to 1 indicate slow mixing and the spatial configuration does not change much from generation to generation, while values close to 0 indicate fast mixing and rapid convergence to the equilibrium spatial distribution $\mathbf{v}^*$.^[13] showed how this quantity governs the bias from estimating an aggregate $\mathcal{R}(t)$ for spatially structured populations without a spatially structured approach. In matrix population models^[2], the alternative damping ratio $d(t) = \frac{1}{s(t)} = \frac{\mathcal{R}(t)}{|\lambda_2(t)|}$ is often used instead, while the spectral gap $\zeta = \mathcal{R}(t) - |\lambda_2(t)|$ can also quantify how long transience ensues before asymptotic behaviour occurs (with large values of ζ implying faster convergence to asymptotic behaviour).

We can further explore transient perturbations by defining the reactivity of the NGM^[2]:

$$\sigma(t) := \mathcal{R}(t) \left(\frac{\mathbf{R}(t) + \mathbf{R}(t)^\top}{2} \right) = \|\mathbf{R}(t)\|_2, \quad (22)$$

which is the largest eigenvalue of the symmetric part of the non-negative matrix $\mathbf{R}(t)$, or equivalently the largest singular value/spectral norm of $\mathbf{R}(t)$. $\sigma(t)$ can be greater than 1 even when $\mathcal{R}(t)$ is less than 1, which implies that epidemic, per-generation perturbations can amplify before the eventual asymptotic decay. In other words, there can be transient growth in incident infections, despite asymptotic epidemic decline. This will happen when we have asymmetric mobility patterns and hence, asymmetric $\mathbf{R}(t)$, which we anticipate for most real-world cities. This means we can capture transient dynamics and the effects of perturbations (e.g. interventions or importations). More generally, $\sigma(t) \geq \mathcal{R}(t)$ with equality if and only if $\mathbf{R}(t)$ is a normal matrix (in which case $\mathbf{R}(t)\mathbf{R}(t)^\top = \mathbf{R}(t)^\top\mathbf{R}(t)$). The right singular vector of $\mathbf{R}(t)$ corresponding to the largest singular value will give the largest amplification from infected introductions.

To quantify the duration of transient growth in epidemic dynamics across n generations, using $\mathbf{E}$ as short-hand for the incidence vector across locations, we can define the amplification envelope $A(n)$ ^[2] as

$$A(n) := \|\mathbf{R}(t)^n\|_2 = \max_{\|\mathbf{E}\|_2=1} \|\mathbf{R}(t)^n\mathbf{E}\|_2, \quad (23)$$

as this is the maximum multiplication/amplification factor after n generations, irrespective of the initial incidence. The trivial cases are $A(0) = 1$ and $A(1) = \sigma(t)$ which imply respectively no amplification after 0 generations and that one-generation amplification is just the reactivity of the NGM. For non-normal matrices (i.e. heterogeneous infection, mixing and movement patterns), we can have $A(n) \gg \mathcal{R}(t)^n$ for small n , while normal matrices will have $A(n) = \mathcal{R}(t)^n$ for all n .

3.1.4 Between-location, inward, and meeting-location transmission

To capture how infections are propagating across locations in the network at any given time, we can write between-location renewal equations as:

$$E^{k \rightarrow j}(t, 0) = R^{kj}(t) \int_0^t g^{kj}(t, a_E) \cdot E^k(t - a_E, 0) da_E. \quad (24)$$

For example, within-location incidence could be compared to incidence resulting from residents of other locations, thereby showing the importance of inward human movement to a location.

We can therefore set up a renewal equation matrix $\mathbf{E}(t, 0)$ with individual elements $\mathbf{E}(t, 0)_{jk}$ describing the number of new infections generated by previously infected individuals from location k in location j . Rows thereby sum to the number of new infections occurring in residents of location j at time t , columns sum to the number of new infections

generated by residents of location k at time t .

$$\mathbf{E}(\mathbf{t}, \mathbf{0}) = \begin{pmatrix} E^{1 \rightarrow 1}(t) & E^{1 \rightarrow 2}(t) & \dots & E^{1 \rightarrow \mathcal{L}}(t) \\ E^{2 \rightarrow 1}(t) & E^{2 \rightarrow 2}(t) & \dots & \dots \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & E^{\mathcal{L} \rightarrow \mathcal{L}}(t) \end{pmatrix}$$

The diagonal elements describe the within-location transmission from the renewal equation (similar to a traditional single-location equation, but with inward and outward human population movements accounted for), while other elements describe pairwise-location transmission.

To capture the inward transmissibility pressure, we can define the expected number of new infections occurring in residents of location j at time t by infected individuals resident in any location as:

$$R_{in}^j(t) := \sum_{k \in \mathcal{N}(j)} R^{kj}(t) = \sum_{k \in \mathcal{N}(j)} \int_0^t \mathbb{K}^{kj}(t, a_E) da_E, \quad (25)$$

which can be studied for the usual threshold property of an outbreak in that location, $R_{in}^j(t)$ above/below 1. This retains familiar yet not identical meaning from the closed population setting for each location j as it controls instantaneous growth/decay at time t for the epidemic in residents of location j . In particular, $R_{in}^j(t)$ is the expected number of new human infections that an infected individual *in any location* would generate in residents of location j throughout the course of their infection, if conditions affecting transmission *from each location to residents in location j* were to remain the same. To quantify bias from a single-patch renewal equations approach, $R_{in}^j(t)$ can be compared to the result from independently estimating $R^j(t)$ for each location, as the independent approach will take into account the local residents' infectiousness pressure without modelling the inward and outward population movements at location j throughout the day.

We can also define the inward generation time distributions and inward instantaneous kernels per location j :

$$\mathbb{K}_{in}^j(t, a_E) := \sum_{k \in \mathcal{N}(j)} \mathbb{K}^{kj}(t, a_E), \quad (26)$$

$$g_{in}^j(t, a_E) := \frac{\mathbb{K}_{in}^j(t, a_E)}{R_{in}^j(t)}, \quad (27)$$

which capture respectively, for residents of each location j at time t , how many expected new infections are to occur, and the relative contribution to new infections, resulting from individuals of infection age a_E resident in any location in the network. While $R_{in}^j(t)$ informs the magnitude of interventions required, $g_{in}^j(t, a_E)$ can inform how such long interventions are needed for the residents of location j to experience an outbreak under control.

Decision-makers are often interested in targeting interventions at locations *where* transmission occurs. We can also define kernels, reproduction numbers, generation times, and incidence for meeting locations.

$$\begin{aligned} \mathbb{K}_{meeting}^l(t, a_E) &:= \sum_{j \in \mathcal{N}(k)} \sum_{k \in \mathcal{N}(l)} f^{jl}(t, S) \cdot S^j(t) \cdot f^{kl}(t, E) \lambda_E^{kl}(t, a_E), \\ R_{meeting}^l(t) &:= \int_0^t \sum_{j \in \mathcal{N}(k)} \sum_{k \in \mathcal{N}(l)} f^{jl}(t, S) \cdot S^j(t) \cdot f^{kl}(t, E) \lambda_E^{kl}(t, a_E) da_E = \int_0^t \mathbb{K}_{meeting}^l(t, a_E) da_E, \\ g_{meeting}^l(t, a_E) &:= \frac{\mathbb{K}_{meeting}^l(t, a_E)}{R_{meeting}^l(t)}, \\ E_{meeting}^l(t, 0) &= \int_0^t \sum_{j \in \mathcal{N}(k)} \sum_{k \in \mathcal{N}(l, j)} f^{jl}(t, S) \cdot S^j(t) \cdot f^{kl}(t, E) \lambda_E^{kl}(t, a_E) E^k(t - a_E, 0) da_E. \end{aligned}$$

which can quantify instantaneously at time t for location l respectively: how many infections are expected by individuals of infection age a_E , how many infections are expected over all infected individuals' lifetime should conditions affecting transmission in l remain the same, the relative contributions from individuals of infection age a_E , and how many infections occur at time t . Likewise, to understand effective susceptible $S_{eff}^l(t)$ and infected populations $E_{eff}^l(t)$ at any

	Between-location	Inward	Outward	Meeting	System
Instantaneous kernel	$\mathbb{K}^{kj}(t, a_E)$ How many infections generated by residents of location k of infection age a_E in residents of location j	$\mathbb{K}_{in}^j(t, a_E)$ How many infections generated in residents of location j by individuals of infection age a_E that are resident in any location	$\mathbb{K}_{out}^j(t, a_E)$ How many infections generated by residents of location j of infection age a_E in residents of any location	$\mathbb{K}_{meeting}^j(t, a_E)$ How many infections generated at meeting location j by individuals of infection age a_E that are residents of any location	
Instantaneous reproduction number	$R^{kj}(t)$ How many infections generated by residents of location k in residents of location j	$R_{in}^j(t)$ How many infections generated in residents of location j by individuals that are resident in any location	$R_{out}^j(t)$ How many infections generated by residents of location j in residents of any location	$R_{meeting}^j(t)$ How many infections generated at meeting location j by individuals that are residents of any location	$\mathcal{R}(t)$ Asymptotic, per-generation growth factor for the linearised epidemic on the network
Generation time distribution	$g^{kj}(t, a_E)$ The proportion of new infections at time t in residents of location j that are generated by residents of location k of infection age a_E	$g_{in}^j(t, a_E)$ The proportion of new infections at time t in residents of location j that are generated by residents of any location that are of infection age a_E	$g_{out}^j(t, a_E)$ The proportion of new infections generated (in residents of any location) by residents of location j that are of infection age a_E	$g_{meeting}^j(t, a_E)$ The proportion of new infections generated at meeting location j that are by individuals of infection age a_E and are residents of any location	

Table 1: All of these are instantaneous measures of epidemic dynamics. All counterfactual. Similar quantities can be derived for vector-borne diseases, where the meeting location summed over represents the residence of the mosquito,

location l at any time t (e.g. during-day versus during-night or during-week versus weekend), :

$$S_{\text{eff}}^l(t) := \sum_{k \in \mathcal{N}(l)} f^{jl}(t, S) \cdot S^j(t),$$

$$E_{\text{eff}}^l(t) := \int_0^t \sum_{k \in \mathcal{N}(l)} f^{jl}(t, E) \cdot E^j(t, a_E).$$

3.2 Case study: Application of our framework to epidemics in different cities

3.3 Mobility-informed PDEs, renewal equations and reproduction numbers for vector-borne diseases

Vector-borne diseases, such as dengue, Zika, or lyme disease, require an intermediate vector to spread the pathogen from one human to another human.

Following eq. (2) and^[34], for a mosquito-borne disease, we can set up a similar age-structured system of PDEs across human and mosquito populations in locations j

$$\frac{\partial E^j}{\partial t} + \frac{\partial E^j}{\partial a_E} = -b_{EI}^j(t, a_E) E^j(t, a_E), \quad (28)$$

$$\frac{\partial I^j}{\partial t} + \frac{\partial I^j}{\partial a_I} = 0, \quad (29)$$

$$\frac{\partial W^j}{\partial t} + \frac{\partial W^j}{\partial a_W} = -b_{WV}^j(t, a_W) W^j(t, a_W) - \mu^j(t) W^j(t, a_W), \quad (30)$$

$$\frac{\partial V^j}{\partial t} + \frac{\partial V^j}{\partial a_V} = 0, \quad (31)$$

where E and I denote infected and infectious humans respectively and W and V denote infected and infectious mosquitoes respectively. We close the system with the following boundary conditions that are informed by human mobility :

$$E^j(t, 0) = \sum_{k \in \mathcal{N}(j)} \int_0^\infty b_{VE}^{kj}(t, a_V) V^k(t, a_V) da_V, \quad (32)$$

$$= \sum_{k \in \mathcal{N}(j)} \int_0^\infty b^k(t) f^{jk}(t, S_H) \frac{S_H^j(t)}{\sum_{l \in \mathcal{N}(k)} f^{lk}(t, S_H) N_H^l(t)} V^k(t, a_V) da_V, \quad (33)$$

$$E^j(0, a_E) = p_E^j(a_E), \quad (34)$$

$$I^j(t, 0) = \int_0^\infty b_{EI}^j(t, a_E) E^j(t, a_E) da_E, \quad (35)$$

$$I^j(0, a_I) = p_I^j(a_I), \quad (36)$$

$$W^j(t, 0) = \sum_{k \in \mathcal{N}(j)} \int_0^\infty b_{IW}^{kj}(t, a_I) I^k(t, a_I) da_I, \quad (37)$$

$$= \sum_{k \in \mathcal{N}(j)} \int_0^\infty \frac{b^k(t) S_M^k(t)}{\sum_{l \in \mathcal{N}(k)} f^{lk}(t, I) N_H^l(t)} f^{kj}(t, I) I^k(t, a_I) p_{WI}(a_I) da_I \quad (38)$$

$$W^j(0, a_W) = p_W^j(a_W), \quad (39)$$

$$V^j(t, 0) = \int_0^\infty b_{WV}^j(t, a_W) W^j(t, a_W) da_W, \quad (40)$$

$$V^j(0, a_V) = p_V^j(a_V). \quad (41)$$

where $b_{VE}^{kj}(t, a_V)$ and $b_{IW}^{kj}(t, a_I)$ represent the infection processes by which a susceptible human and mosquito resident in location j respectively are infected by an infectious human and mosquito respectively. The development processes $b_{EI}^j(t, a_I)$ and $b_{WV}^j(t, a_V)$ represent the rate at which infected humans and mosquitoes become infectious. $p_E^j(a_E)$, $p_I^j(a_I)$, $p_W^j(a_W)$, and $p_V^j(a_V)$ are the initial densities for the resident exposed human, infectious human, exposed mosquito, and infectious mosquito populations respectively. We have assumed that locations are sufficiently large in area such that mosquitoes can only infect and be infected within their home location. This is the case for most real-world applications to surveillance data, but to account for between-location movement of mosquitoes, this would simply require an analogous term to $f^{jk}(t, S_H)$ for mosquitoes that represents the probability that each mosquito resident in location j is in location k at any time t and then modified effective mosquito population terms in renewal equations for the possible meeting locations with susceptible and infected humans.

For each locations' birth/development processes, we can follow^[34] and write these as:

$$\begin{aligned} b_{EI}^j(t, a_E) &= \lambda_{E, \text{Latent}}^j(a_E) \\ b_{IW}^j(t, a_I) &= \frac{b^j(t)}{N_H^j(t)} \cdot S_M^j(t) \cdot p_{WI}(a_I) \\ b_{WV}^j(t, a_W) &= \lambda_{W, \text{Latent}}^j(t, a_W) \\ b_{VE}^j(t, a_V) &= b^j(t) \cdot \frac{S_H^j(t)}{N_H^j(t)} \cdot p_{EV}(a_V) \end{aligned} \quad (42)$$

where $\lambda_{E, \text{Latent}}^j(a_E)$ and $\lambda_{W, \text{Latent}}^j(t, a_W)$ denote the rates at which latent periods end for infected humans and mosquitoes respectively, $N_H^j(t)$ and $N_M^j(t)$ denote the total human and mosquito populations, $S_H^j(t)$ and $S_M^j(t)$ denote the susceptible human and mosquito population counts, p_{WI} and p_{EV} denote the probability of mosquito and human infection upon infectious bite respectively, and $b^j(t)$ denotes the rate at which a mosquito bites a human.

Hazard rates for the human latent periods $b_{EI}^j(t, a_E)$ may differ by population demographics (e.g. older populations), while the mosquito latent period $b_{WV}^j(t, a_W)$ may differ by location/space (e.g. urban landscape and mosquito adaptation) in addition to time.

Following the method of characteristics and seeking a separable solution, the renewal equation for new human infections in residents of location j can be written as:

$$\begin{aligned}
E^j(t, 0) &= \sum_{k \in \mathcal{N}(j)} \int_0^\infty b_{VE}^{kj}(t, a_V) V^k(t, a_V) da_V, \\
&= \sum_{k \in \mathcal{N}(j)} \int_0^\infty b_{VE}^{kj}(t, a_V) V^k(t - a_V, 0) e^{-\mu a_V} da_V, \\
&= \sum_{k \in \mathcal{N}(j)} \int_0^\infty b_{VE}^{kj}(t, a_V) e^{-\mu a_V} \int_0^\infty b_{WV}^j(t, a_W) W^k(t - a_V, a_W) da_W da_V, \\
&= \sum_{k \in \mathcal{N}(j)} \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} b^k(t) f^{jk}(t) \frac{S_H^j(t)}{\sum_{l \in \mathcal{N}(k)} f^{lk}(t) N_H^k(t)} p_{EV}(a_V) b_{WV}^k(t - a_V, a_W) \\
&\quad \cdot \sum_{c \in \mathcal{N}(k)} \frac{b^c(t - a_V - a_W) S_M^c(t - a_V - a_W)}{\sum_{d \in \mathcal{N}(c)} f^{dc}(t - a_V - a_W) N_H^d(t - a_V - a_W)} f^{kj}(t - a_V - a_W) p_{WI}(a_I) \\
&\quad \cdot b_{EI}^c(t - a_V - a_W - a_I, a_E) e^{\int_0^{a_V} -\mu^c(t-a_V-a_W+r)dr} e^{-\int_0^{a_W} b_{WV}^c(t-a_V-a_W+r, r) - \mu^c(t-a_V-a_W+r)dr} \\
&\quad \cdot E^c(t - a_V - a_W - a_I - a_E, 0) e^{-\int_0^{a_E} b_{EI}^c(t-a_V-a_W-a_I-a_E+s, s)ds} da_E da_I da_W da_V
\end{aligned} \tag{43}$$

or compactly:

$$\begin{aligned}
E^j(t, 0) &= \sum_{k \in \mathcal{N}(j)} \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} \sum_{c \in \mathcal{N}(k)} \mathbb{K}^{ckj}(t, a_E, a_I, a_W, a_V) \\
&\quad \cdot E^c(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V
\end{aligned} \tag{44}$$

$\mathbb{K}^{ckj}(t, a_E, a_W, a_I, a_V)$ is the instantaneous rate at which previously infected resident humans of location c produce secondary infections that occur at meeting location k in residents of location j . In further detail, this is the expected number of secondary infections occurring in residents of location j at calendar time t from the bite of infectious mosquitoes resident in location k which became infectious at calendar time $t - a_V$ and were infected at calendar time $t - a_V - a_W$ in location k by an infectious human resident in location c who was originally infected at time $t - a_V - a_W - a_I - a_E$. The ordering of notation (c, k, j) reflects the resident locations of the ordered infection events (primary human infection of resident of location c , infection of mosquito resident of location k , and secondary human infection of resident of location j). As in^[34], the ages in $\mathbb{K}^{ckj}(t, a_E, a_W, a_I, a_V)$ reflect the ages of the individual humans/mosquitoes at the times when they play their role in the multi-stage transmission cycle.

As before, we can sum over meeting (/mosquito resident) locations to write instantaneous kernels that describes instantaneous transmission between humans of two locations:

$$\mathbb{K}^{cj}(t, a_E, a_I, a_W, a_V) = \sum_{k \in \mathcal{N}(j)} \mathbb{K}^{ckj}(t, a_E, a_I, a_W, a_V) da_E da_I da_W da_V, \tag{45}$$

where the expression sums over all meeting locations k where the susceptible human resident in location j can be infected by an infectious mosquito resident in any location k who was themselves infected by the infected human resident in location c .

Different types of human-to-human instantaneous reproduction numbers can also be formed by integrating over all ages. We define the between-location $R^{cj}(t)$, the expected number of secondary human infections in location j generated by a previously infected human resident in location c throughout their infected lifetime if conditions affecting this human transmission route were to remain the same at time t :

$$R^{cj}(t) = \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} \sum_{k \in \mathcal{N}(j)} \mathbb{K}^{ckj}(t, a_E, a_I, a_W, a_V). \tag{46}$$

This suggests between- generation time distributions can be formed as:

$$g^{cj}(t, a_E, a_I, a_W, a_V) = \frac{\mathbb{K}^{cj}(t, a_E, a_I, a_W, a_V)}{R^{cj}(t)}, \tag{47}$$

which describes, at each time t , the relative contributions of different stage-specific ages to the expected number of new human infections for residents of location j arising from the original human infection from a resident of location c . Again, ages are evaluated at the calendar times when each stage-specific individual played their role in the transmission cycle.

Our renewal equation can be written as:

$$\begin{aligned} E^j(t, 0) &= \sum_{c \in \mathcal{N}(k)} \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} \mathbb{K}^{cj}(t, a_E, a_I, a_W, a_V) \\ &\quad \cdot E^c(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V \\ &= \sum_{c \in \mathcal{N}(k)} R^{cj}(t) \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} g^{cj}(t, a_E, a_I, a_W, a_V) \\ &\quad \cdot E^c(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V \end{aligned} \quad (48)$$

The per-location outward instantaneous kernels

$$\mathbb{K}_{out}^c(t, a_E, a_I, a_W, a_V) = \sum_{j \in \mathcal{N}(c)} \sum_{k \in \mathcal{N}(j)} \mathbb{K}^{ckj}(t, a_E, a_I, a_W, a_V) da_E da_I da_W da_V, \quad (49)$$

which suggests defining an outward instantaneous reproduction number $R_{out}^c(t)$ as:

$$\begin{aligned} R_{out}^c(t) &= \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} \mathbb{K}_{out}^c(t, a_E, a_I, a_W, a_V) \\ &= \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} \sum_{j \in \mathcal{N}(c)} \sum_{k \in \mathcal{N}(j)} \mathbb{K}^{ckj}(t, a_E, a_I, a_W, a_V) da_E da_I da_W da_V \end{aligned} \quad (50)$$

This sums over all locations j where a secondary infected human could be resident to capture, at time t , how many expected infections are generated by the infected individual resident in location c , if conditions affecting human transmission were to remain the same. Note we have also summed over all possible locations where the intermediate mosquito was infected, or equivalently where the infected human resident in location j was infected.

$$g_{out}^c(t, a_E, a_I, a_W, a_V) = \frac{\mathbb{K}_{out}^c(t, a_E, a_I, a_W, a_V)}{R_{out}^c(t)}, \quad (51)$$

which describes, at each time t , the relative contributions of different stage-specific ages to the expected number of new human infections arising from the original human infection from a resident of location c .

We can define the expected number of new infections occurring in residents of location j by infected individuals resident in any location throughout their infected lifetime if transmission conditions were to remain the same.

$$R_{in}^j(t) = \sum_{k \in \mathcal{N}(j)} \sum_{c \in \mathcal{N}(k)} \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} \mathbb{K}^{ckj}(t, a_E, a_I, a_W, a_V) da_E da_I da_W da_V, \quad (52)$$

which indicates whether a location is experiencing net instantaneous growth/decay, and whether it is a sink of transmission.

The different formulations of R , g , and K all count different classes of infections and therefore define differently the expected numbers of new infections, the times between infections, and the expected contributions of different ages to new ages.

We can also compute between-location renewal equations

$$E^{c \rightarrow j}(t, 0) = R^{cj}(t) \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} g^{cj}(t, a_E, a_I, a_W, a_V) \cdot E^c(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V \quad (53)$$

As before, we can set up renewal equation matrix $\mathbf{E}(t, 0)$ with individual elements $\mathbf{E}(t, 0)_{jc} = E^{c \rightarrow j}(t, 0)$ describing the number of new human infections generated by previously infected individuals from location c in location j .

In matrix form,

$$\mathbf{E}(t, \mathbf{0}) = \begin{pmatrix} R^{11}(t) \int \int \int \int g^{11}(t, a_E, a_I, a_W, a_V) \cdot E^1(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V & E^{2 \rightarrow 1}(t, 0) & \dots & E^{N \rightarrow 1}(t, 0) \\ R^{12}(t) \int \int \int \int g^{12}(t, a_E, a_I, a_W, a_V) \cdot E^1(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V & E^{1 \rightarrow 2}(t, 0) & \dots & E^{N \rightarrow 2}(t, 0) \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & E^{N \rightarrow N}(t, 0) \end{pmatrix}$$

where for simplicity, we have dropped the limits of integration from eq. (53). Rows sum to the number of new infections occurring in location j at time t . This formulation makes explicit the relative contributions of each location c to the new human infections in location j at time t . As before, the diagonal elements describe the within-location transmission from the renewal equation.

We can define the aggregate number of new infections using an across-location renewal equation:

$$|E(t, 0)| = \sum_{j=1}^N \sum_{c \in \mathcal{N}(k)} R^{cj}(t) \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} g^{cj}(t, a_E, a_I, a_W, a_V) \cdot E^c(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V, \quad (54)$$

which is the grand sum of the renewal equation matrix.

To make explicit the row sum calculation that gives rise to the expected number of new infections occurring in human residents of location j at time t , we write:

$$E^j(t, 0) = \sum_{c \in \mathcal{N}(k)} R^{cj}(t) \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} g^{cj}(t, a_E, a_I, a_W, a_V) \cdot E^c(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V \quad (55)$$

3.4 Resulting dynamics; GT, $\mathbf{R}(t)$, Incidence

Figure 2: Simulated data and overall outputs

- A) Mobility across space and time (3-D or two tile plots)
- B) Incidence across space and time (tile plot) and total incidence
- D) System-level $\mathcal{R}(t)$
- E) Elasticity of $\mathcal{R}(t)$ over time wrt human movement proportions and total elasticity per location over time (i.e. which locations most important for network growth)

Figure 3: Spatiotemporal trends

- A) $R_{out}^k(t)$, including threshold line at 1
- B) $R_{in}^k(t)$
- C) 3-D Plot of $R^{kj}(t)$ and snapshots (tile plot) around early, peak, and late
- D) Independent $\mathbf{R}(t)$ vs $R_{in}^k(t)$.
- E) Generation time comparison. Overlaid density plots at a snapshot time showing: the biological input, the within-location for a central versus peripheral location, a between-location for a close pair and for a distant pair, and the network-level generation time.
- F) Spatial outputs over time: Fraction of incidence within vs outside, reactivity vs spectral radius, and mixing time ratio. Highlight where reactivity greater than one unlike $\mathbf{R}(t)$ is not.

Figure 4: Different commuting pattern

- A) Mobility
- B) 3-D Plot of $R^{kj}(t)$
- C) Spatial outputs over time: Fraction of incidence within vs outside, reactivity vs spectral radius, and mixing time ratio
- D) Elasticity of $\mathcal{R}(t)$ over time wrt human movement proportions
- E) Counterfactual predictions: Reduce outward flows of top 10% locations outward $\mathbf{R}(t)$ (exporters), reduce inward flows of top 10% locations inward (hubs),

SI Figure 1: Further outputs

- A) Generation times between locations vs local
- B) Between-location $R(t)$ sensitivity
- C) Inward and outward $R(t)$ sensitivity

SI Figure 1: Further sensitivity metrics

- A) Generation time sensitivity
- B) Between-location $R(t)$ sensitivity
- C) Inward and outward $R(t)$ sensitivity

SI Figure 2: Further network characteristics and their impacts

- A) Homogeneous reductions by amount vs targeted reductions by same amount.
- B) Counterfactual infectiousness profile (e.g. non-SARS-CoV-2)
- C) Counterfactual contact rate (e.g. information campaign)

SI Figure 2: Further counterfactual predictions

- A) Homogeneous reductions by amount vs targeted reductions by same amount.
- B) Counterfactual infectiousness profile (e.g. non-SARS-CoV-2)
- C) Counterfactual contact rate (e.g. information campaign)

Notes (to be removed): Many things we can explore.....

- Effect of various GT, R_t ,
- Elasticity of $R(t)$
- Which features (e.g. length of stay, distance, number of trips) matter most? How varies by city?
- Which patches most influential? How varies by commuting patterns, time?
- High-degree nodes effects etc

- Effects: Time-varying + overall sensitivity (which vars most sensitive to, i.e. which assumptions most important to get right), overall epidemic size etc e.g. wrt

- within-location time fraction and within-neighbor
- cross-location fraction
- commuting duration
- long-distance flows

Repeat for system-level, inward- and outward-location, between-location $R(t)$ and incidence (peak, time-varying, aggregate)

Controllability: Can operate on restricted set of nodes.

Advantage is have causally explicit model that does not ignore external influences on a location's outbreak. Quantify:

- Elasticity: % reductions in $R(t)$, peak, length of outbreak by different mobility interventions

4 Discussion

So what? Introduce a novel, mechanistic framework with clear policy information about targeting interventions across space and time during an outbreak.

that addresses a fundamental spatial limitation of most common way to estimate $R(t)$ as extend familiar renewal equation to allow for movement of humans (both infected and susceptible) across locations. This allows us to examine source-sink dynamics across mobility networks and capture how epidemics can be brought under control by targeted interventions. Our framework captures the wide array of within-day infection opportunities, created by the amount of time that individuals spend in their home location and other locations.

This framework introduces many new, policy-relevant quantities (Table 1) that are not obtainable from single-location renewal equation. This poses the question; which reproduction number should be used for informing epidemic control strategies? We argue that this depends on the policy question as i) between-location reproduction numbers can inform temporary, targeted reductions in specific between-location mobility flows, ii) inward reproduction numbers can inform targeted interventions (e.g. reduce inward flow) for locations that are experiencing instantaneous net growth in the number of individuals being infected, iii) outward reproduction numbers can inform targeted interventions for the outward flow of residents from locations that are contributing (instantaneously), on average, more than one secondary infection, iv) meeting-location reproduction numbers can inform targeted interventions for locations where growing totals of infections are occurring, and finally v) system-level reproduction numbers can inform interventions targeted at the network as a whole. These reproduction numbers inform both the timing and magnitude of interventions needed for controlling different aspects of the outbreak. Analogous quantities (e.g. pairwise, inward etc.) for the generation time distributions are also important, not just for estimating $R(t)$ more reliably, but also for knowing *when* and *how fast* infections will occur (not just magnitude), the type of intervention that might be suitable, and how long that intervention might be needed. As usual, estimating time-varying growth rates $r(t)$ can provide complementary information on how fast infections are occurring within and between locations. Recognising that instantaneous measures can be misleading^[2], we further adapt existing measures of transience to our framework. This is important for understanding how short-term growth in the incidence of new infections can be generated from importation events across locations (and possible short-term interventions may be needed), even when $\mathcal{R}(t)$ indicates asymptotic equilibrium decay.

Previous studies: Several methods have been developed to estimate $R(t)$ across space^[1]. However, these per-location $R(t)$ estimates differ to the quantities above and often do not have a mechanistic meaning, due to either spatial smoothing or assuming that inward and outward transmission occurs according to the same reproduction process. For example, such approaches allow residents from location k to generate infections in residents of a location j , yet the $R(t)$ estimate for location j is used for the reproduction process from all residents of other locations k into residents of location j . Metapopulation models exist for mosquito-borne diseases (e.g. ^[2]), but these are not linked to instantaneous, location-specific measures of human-to-human transmissibility.

Type of human mobility data matters...type of human movement matters; distance, length of time, all of which important. Many studies have performed association studies where the effects of mobility on transmission are inferred/deduced using association between mobility and $R(t)$. However, this approach abstracts away the true causal mechanisms by which human movement shapes epidemics. In future studies, counterfactual predictions (a gold standard for causal inference) using different human movement patterns could be obtained using our causally explicit, mechanistic framework. This is not just a statistical issue, but also a trust issue for modelling and public health policy as inferred effects and information for decision-makers are not confounded by external factors.

Inference + future directions: The unveiled importance of human movement is especially relevant for inference of $R(t)$, which often feeds into real-time analyses for informing decision-making. How would inference work? First, conditional on having case incidence data, our framework would require only one additional data stream (compared to existing methods), that is high-quality, relevant data/estimates on human mobility that capture within- and across-day movements of individuals in the network. Shown biases of estimating independently, yet identifiability will be an issue, particularly in resource-limited settings where human movement patterns are poorly observed. This could motivate transfer learning where mobility data from other settings could be paired with a calibration model, as well as the next generation of mobility models

Then, these data could be embedded into our current mechanistic formula alongside existing joint inference techniques (e.g. Bayesian state-space models, akin to existing software such as EpiFilter or EpiNow2), with the additional complexity of multivariate latent states which will require careful considerations of scalability.

ABMs: Probabilistic

Further perturbation analyses, further network analyses and alternative network models^[42] Epidemic control:^[16,43]. We could also derive the type-reproduction number $T_j(t)$ for location j at time t using the NGM, as this would possess the threshold property and explicitly accounts for all direct and indirect transmission chains that return to location j . blending techniques from controlling complex networks^[2]

Practical data collection + extensions: Current application is limited by availability of real-world human movement data that captures fraction of time spent in different locations over time. Our study was not focused on modelling real-world city patterns, but rather if we had access to such data, what is feasible. Ill-specified f - our sensitivity metrics allow us to capture where information most valuable. Mobility approximations may be valuable, possibly based on origin-flow data paired with assumptions or data (e.g. from censuses) on commuter stay durations. This will be important as different mobility patterns will give rise to different epidemic patterns across space and time^[2], and likely suggest different types and strengths of interventions

Formulation of f_{ij} allows for long-term/distance migration, while not explicitly considered.

So-what again? within-day infection opportunities.

Notes: Demonstrate t - Human mobility importance, policy implications - Type of human mobility data matters... - Similarly, type of human movement matters; distance, length of time, - Pathogen types/transmission routes -

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Glossary

Term	Description
u_{lim}	Daily vaccination capacity (doses per day).
U_{lim}	Total vaccine stockpile available for the duration of the epidemic.
$\eta(t)$	Shadow price of the daily capacity constraint; represents the marginal cost of a one-unit increase in u_{lim} .
$j(t)$	Instantaneous incidence (rate of new infections per unit time).
$C(t)$	Cumulative incidence; the total number of individuals infected since $t = 0$.
ϵ	Vaccine inefficacy ($1 - \text{efficacy}$).

Notes

Notes/thoughts: Will remove once incorporated

- Derive for directly transmitted + VBD - done (see scribbles, mob_Derivations + mre_mob_derivations)
- Simulations
- Repeat independently and compare to jointly, show bias from independent fits and from using flow data, synchrony,
- Mobility sensitivity and elasticity indices — percent change in metric (e.g. $R(t)$, peak size, length of season) per % change in mobility. How does sensitivity depend on where you are in outbreak and where in the network?.

Epidemic speed — the spatial spreading rate; i.e. the average spatial generation distance — how far, on average, a secondary infection is from its primary infection. Combined with the mean generation time, the ratio gives estimate of the epidemic’s spatial velocity: $v(t) = \frac{\bar{d}(t)}{\bar{g}(t)}$

Adding an incubation period is rather simple, but does require an extra state variable I . Repeating procedure...

New quantities

Will write down theorem + conditions for $\mathbf{R}(t)$ here

A Technical conditions and eigenvalue problem for reproduction number definition

The Perron-Frobenius theorem guarantees that:

1. $\mathcal{R}(t) = \rho(\mathbf{R}(t))$ is a simple, real, positive eigenvalue that is strictly larger than all other eigenvalues
2. There is a unique, right eigenvector $\mathbf{v}^*$ such that $\mathbf{R}\mathbf{v}^* = \mathcal{R}(t)\mathbf{v}^*$.
3. There is also a unique left eigenvector $\mathbf{v}$ such that $\mathbf{v}\mathbf{R} = \mathcal{R}(t)\mathbf{v}^T$.

By normalising such that $\sum_{j=1}^{\mathcal{L}} v_j^* = 1$, the right eigenvector defines a probability distribution that describes the stable/equilibrium, long-term spatial distribution of infections across locations. In words, this means that irrespective of the initial conditions for the incidence spatial distribution, after many generations with a counterfactually constant $\mathbf{R}(t)$, the fraction of all infections occurring in location k converges to v_k^* . Meanwhile, the left eigenvector entries v_j measure the contribution of a single infected individual in location j .

The instantaneous kernel $K(t, \tau)$ satisfies $K(t, \tau) \geq 0$ as